AP CALCULUS AB

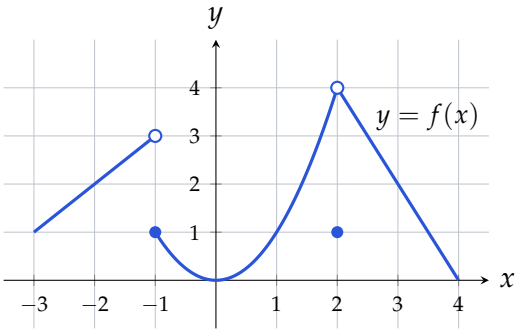
Unit 1 • Lesson 1.9 practice

Lesson 1.9 • Connecting Multiple Representations of Limits

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.



The graph of f: a line segment for $-3 \le x < -1$, $y = x^2$ for $-1 \le x < 2$, a line segment for $2 < x \le 4$; $f(-1) = 1$, $f(2) = 1$.

Question 1

NO CALCULATOR

Let $g(x) = x + 1$. Using the graph of f, what is $\lim_{x \to 2} f(x)g(x)$?

(A) 3
(B) 4
(C) 12
(D) 7

Question 2

NO CALCULATOR

With $g(x) = x + 1$, what is $\lim_{x \to 2} \big(f(x) - g(x)\big)$?

(A) 1
(B) -2
(C) -1
(D) 7

Question 3

NO CALCULATOR

A function h has values 4.9, 4.99, 4.999, 5.001, 5.01, 5.1 at $x = 0.9, 0.99, 0.999, 1.001, 1.01, 1.1$. Let $k(x) = 2x$. What is $\lim_{x \to 1} \dfrac{h(x)}{k(x)}$?

(A) 5
(B) $\frac{5}{2}$
(C) 10
(D) $\frac{2}{5}$

Question 4 NO CALCULATOR

With $g(x) = x + 1$, what is $\lim_{x \to 3} g(f(x))$?

(A) 4
(B) 2
(C) 5
(D) 3

Question 5 (graph) NO CALCULATOR

Use the graph of f above. What is $\lim_{x \to -1^-} x\,f(x)$?

(A) -1
(B) 3
(C) -3
(D) It does not exist.

Question 6 (Free response) NO CALCULATOR

Use the graph of f above. The function g is defined for all x, and $g(1.9) = -1.2$, $g(1.99) = -1.02$, $g(1.999) = -1.002$, $g(2.001) = -0.998$, $g(2.01) = -0.98$, $g(2.1) = -0.8$.

(a) Assuming the table shows the trend of g, find $\lim_{x \to 2} \big(f(x) + g(x)\big)$.
(b) Find $\lim_{x \to 2} f(x)g(x)$ and $\lim_{x \to 2} \frac{f(x)}{g(x)}$.
(c) Find $\lim_{x \to -1^+} \big(f(x) + x\big)$ and $\lim_{x \to -1^-} \big(f(x) + x\big)$.